Analysis of Transistor-level CMOS Inverter in LTspice

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1 Design Parameters & Tool Setup

1.1 Parameters

The NMOS and PMOS transistors are modeled using the typical values for 180nm CMOS Small-Signal Model [1]. The SPICE directives for the NMOS and PMOS transistors are as follows:

- **NMOS:** `.model NMOS_180nm NMOS (LEVEL=1 VTO=0.8 KP=90u GAMMA=0.5 PHI=0.68 LAMBDA=0.06)`
- **PMOS:** `.model PMOS_180nm PMOS (LEVEL=1 VTO=-0.9 KP=30u GAMMA=0.5 PHI=0.68 LAMBDA=0.06)`

1.2 LTspice Schematic

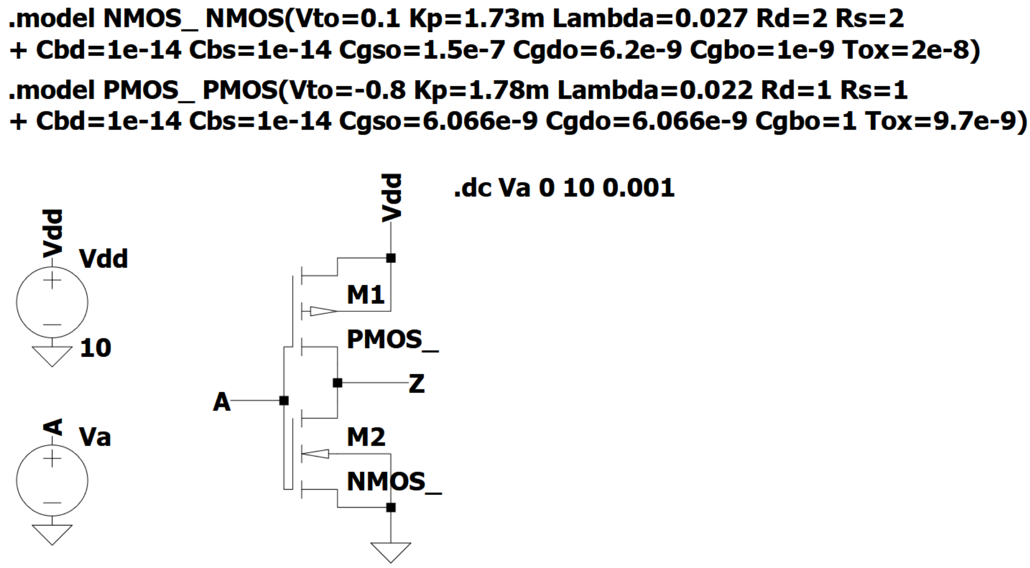

Figure 1: Circuit setup of the transistor-level CMOS inverter in LTspice.

From Fig. 1, the circuit setup of the transistor-level CMOS inverter in LTspice is shown. First, on the left side, there are two voltage sources, the power supply voltage (Vdd) is set to 10V , while V2 represents the input voltage (A). Then, on the right side, there are two MOSFETs used to represent the PMOS and NMOS transistors. The upper side is the PMOS transistor, which is connected to Vdd, while the lower side is the NMOS transistor, which is connected to the ground. Finally, the gate of both transistors is connected to A and the output voltage (Z) is connected to the junction between the two transistors. [1]

2 DC Analysis (Voltage Transfer Characteristics)

2.1 VTC Curve Generation

To obtain the Voltage Transfer Characteristic (VTC), a DC sweep simulation (`.dc`) is performed, sweeping the input voltage from 0V to [Vdd] with a step size of [Step Size].

[Insert VTC Curve Graph Here - Uncomment the includegraphics above when ready]

Figure 2: Voltage Transfer Characteristic (VTC) of the CMOS Inverter.

2.2 Critical Voltages

From the VTC curve, the critical voltages are extracted (typically where the slope is -1):

- V_{OH} = [Result] V
- V_{OL} = [Result] V
- V_{IH} = [Result] V
- V_{IL} = [Result] V

2.3 Noise Margins

The noise margins are calculated using the extracted critical voltages:

- High Noise Margin ($NM_H = V_{OH} - V_{IH}$) = [Result] V
- Low Noise Margin ($NM_L = V_{IL} - V_{OL}$) = [Result] V

2.4 Switching Threshold

The switching threshold (V_M), where $V_{in} = V_{out}$, is measured from the VTC:

- V_M = [Result] V

3 Transient Analysis (Timing Characteristics)

3.1 Transient Simulation Setup

To analyze the timing characteristics, a transient simulation (`.tran`) is performed over [Simulation Time]. The input is driven by a pulse voltage source with the following parameters:

- $V_{initial}$ = [0] V, V_{on} = [Vdd] V
- T_{delay} = [Delay], T_{rise} = [Rise], T_{fall} = [Fall]
- T_{on} = [Pulse Width], T_{period} = [Period]

[Insert Transient Waveforms (Input and Output) Graph Here]

Figure 3: Transient response of the CMOS Inverter.

3.2 Propagation Delay

The propagation delay is measured between the 50% points of the input and output transitions:

- High-to-Low Delay (t_{pHL}) = [Result]
- Low-to-High Delay (t_{pLH}) = [Result]
- Average Delay (t_p) = [Result]

3.3 Rise and Fall Times

The rise and fall times are measured between the 10% and 90% points of the output signal:

- Rise Time (t_r) = [Result]
- Fall Time (t_f) = [Result]

4 Power Analysis

4.1 Dynamic Power

Dynamic power dissipation is measured during the transient simulation:

- Average Dynamic Power ($P_{dynamic}$) = [Result]

4.2 Static Power

Static (leakage) power is calculated by measuring the current drawn from Vdd when the inverter state is steady:

- Input Low ($V_{in} = 0V$), P_{static} = [Result]
- Input High ($V_{in} = Vdd$), P_{static} = [Result]

References

[1] Scribd. 180nm process typical parameter values. https://www.scribd.com/presentation/470685433/180nm-process-typical-parameter-values.